

1. MMF

$$\Delta T = \frac{(n_{\text{core}} - n_{\text{clad}})^2}{8cn_{\text{core}}} L$$

$$L = \frac{\Delta T 8cn_{\text{core}}}{(n_{\text{core}} - n_{\text{clad}})^2}$$

For a modal dispersion of 10% $T_b \Rightarrow \Delta T = 10\text{ps}$ for $T_b = 100\text{ps}$

$$L = \frac{(10\text{ps})(8)(3 \times 10^8 \text{ m/s})(1.48)}{(1.48 - 1.46)^2} = 88.8 \text{ m}$$

$$L = 88.8 \text{ m}$$

2. SMF

12

$$10 \text{Gb/s} \quad w/ \lambda = 1310 \text{nm} \quad \Delta\lambda = 3 \text{nm} \quad \bar{P} = 1 \text{mW}$$

$$\text{SMF} : \text{loss} = 0.4 \text{dB/km} \quad D = 0.5 \frac{\text{ps}}{\text{nmkm}} \quad \bar{P}_{\text{sens}} = -15 \text{dBm}$$

a. Max Transmission Distance?

$$* \text{ Loss Limit} \Rightarrow \frac{\text{TX Power}}{(\text{Fiber Loss})(\text{RX Sens})}$$

$$\frac{\text{TX Power}}{\text{RX Sens}} = 0 \text{dBm} - (-15 \text{dBm}) = 15 \text{dBm}$$

$$L = \frac{15 \text{dBm}}{0.4 \text{dB/km}} = 37.5 \text{km}$$

$$* \text{ Dispersion Limit} \Rightarrow \Delta T \leq \frac{1}{2B} = \frac{1}{2(10 \text{Gb/s})} = 50 \text{ps}$$

$$\Delta T = D \Delta\lambda L$$

$$L = \frac{\Delta T}{D \Delta\lambda} = \frac{50 \text{ps}}{(0.5 \frac{\text{ps}}{\text{nmkm}})(3 \text{nm})} = 33.3 \text{km}$$

$$\boxed{\text{Max } L = 33.3 \text{km}, \text{ Dispersion Limited}}$$

* Note if we include the 1dB dispersion penalty, the max length = 35km w/ $\bar{P}_{\text{sens}} = -14 \text{dBm}$. Thus, we reach the dispersion limit before this.

b. $L = 5 \text{km}$

$$\Delta T = D \Delta\lambda L = \left(\frac{0.5 \text{ps}}{\text{nmkm}} \right) (3 \text{nm}) (5 \text{km}) = 7.5 \text{ps}$$

$$B = \frac{1}{2\Delta T} = \frac{1}{2(7.5 \text{ps})} = 66.7 \text{Gb/s}$$

$$\boxed{B = 66.7 \text{Gb/s}}$$

3. Vertical p-i-n Detector

$$W = 2\mu\text{m} \quad \alpha = 10^4 \text{cm}^{-1} \quad V_n = 10^5 \text{m/s}$$

$$R_{PD} = 20\Omega \quad C_{PD} = 150\text{fF}$$

a. $R = ? \quad \lambda = 1310\text{nm}$

$$R = \eta \frac{q}{hc} \lambda$$

$$\eta = 1 - e^{-\alpha W} = 1 - e^{-(10^4 \text{cm}^{-1})(2\mu\text{m})} = 0.865$$

$$R = (0.865) \left(8 \times 10^5 \frac{\text{A}}{\text{Wm}} \right) (1310\text{nm}) = 0.907 \frac{\text{A}}{\text{W}}$$

b. $BW = ?$

$$\text{Total BW} = \frac{1}{2\pi} \left(\frac{1}{\frac{W}{V_n} + R_{PD} C_{PD}} \right) = \frac{1}{2\pi} \left(\frac{1}{\frac{2\mu\text{m}}{10^5 \text{m/s}} + (20\Omega)(150\text{fF})} \right)$$

$$\text{Total BW} = 6.92\text{GHz}$$

4. Waveguide PD

$$W = 320 \text{ nm} \quad \alpha = 10^3 \text{ cm}^{-1} \quad \text{abs} \quad L = 10 \mu\text{m}$$

$$V_n = 10^5 \text{ m/s} \quad R_{PD} = 50 \Omega \quad C_{PD} = 7 \text{ fF}$$

a. $R = ? \quad \lambda = 1310 \text{ nm}$

$$R = \eta \frac{q}{h c} \lambda$$

$$\eta = 1 - e^{-\alpha L_{\text{abs}}} = 1 - e^{-(10^3 \text{ cm}^{-1})(10 \mu\text{m})} = 0.632$$

$$R = (0.632) \left(8 \times 10^5 \frac{\text{A}}{\text{W}} \right) (1310 \text{ nm}) = 0.662 \text{ A/W}$$

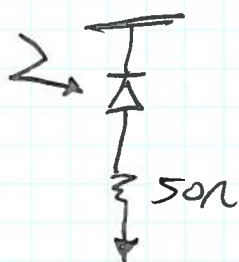
b. $BW = ?$

$$\text{Total BW} = \frac{1}{2\pi} \left(\frac{1}{\frac{W}{V_n} + R_{PD} C_{PD}} \right) = \frac{1}{2\pi} \left[\frac{1}{\frac{320 \text{ nm}}{10^5 \text{ m/s}} + (50 \Omega)(7 \text{ fF})} \right]$$

$$\text{Total BW} = 44.9 \text{ GHz}$$

5. R-Front-End Sensitivity for $BER = 10^{-12}$

15.



$$i_{n,amp}^{rms} = \sqrt{\frac{4KT}{R} BW_0} = \sqrt{\frac{4(1.38 \times 10^{-23})(300)}{50} (106 \text{ Hz})}$$

$$i_{n,amp}^{rms} = 1.82 \mu A$$

a. p-i-n $w/R = 1 \text{ A/W}$

$$\bar{P}_{sens, PIN} = \frac{Q i_{n,amp}^{rms}}{R} + \frac{Q^2 q BW_1}{R}$$

$$= \frac{(7.035)(1.82 \mu A)}{1 \text{ A/W}} + \frac{(7.035)^2 (1.6 \times 10^{-19})(106 \text{ Hz})}{1 \text{ A/W}} = 12.9 \mu W$$

$$\bar{P}_{sens, PIN} = 12.9 \mu W = -18.9 \text{ dBm}$$

$$i_{n, PIN, 1}^{rms} = \sqrt{4q(1 \text{ A/W})(12.9 \mu W)(106 \text{ Hz})} = 287 \text{ nA}$$

$$i_{n, PIN, 1}^{rms} = 287 \text{ nA}$$

b. APD $w/R = 1 \text{ A/W}$, $M=8$, $F=4$

$$\bar{P}_{sens, APD} = \frac{1}{M} \frac{Q i_{n,amp}^{rms}}{R} + F \frac{Q^2 q BW_1}{R}$$

$$= \frac{1}{8} \frac{(7.035)(1.82 \mu A)}{1 \text{ A/W}} + 4 \frac{(7.035)^2 (1.6 \times 10^{-19})(106 \text{ Hz})}{1 \text{ A/W}} = 1.92 \mu W$$

$$\bar{P}_{sens, APD} = 1.92 \mu W = -27.2 \text{ dBm}$$

$$i_{n, APD, 1}^{rms} = \sqrt{4(8^2)(4)q(1 \text{ A/W})(1.92 \mu W)(106 \text{ Hz})} = 1.77 \mu A$$

$$i_{n, APD, 1}^{rms} = 1.77 \mu A$$

$$c. \bar{P}_{sens,OA} = \frac{1}{G} \frac{Q i_{n,amp}^{rms}}{R} + \eta_F \frac{Q^2 q B W}{R}$$

$$= \frac{1}{50} \frac{(7.035)(1.82 \mu A)}{1 A/W} + 2 \frac{(7.035)^2 (1.6 \times 10^{-19})(10 GHz)}{1 A/W}$$

$$\bar{P}_{sens,OA} = 414 nW = -33.8 dBm$$

$$i_{n,OA,1}^{rms} = \sqrt{2(50)^2(4)q(1 A/W)(414 nW)(10 GHz)} = 3.64 \mu A$$

$$i_{n,OA,1}^{rms} = 3.64 \mu A$$